Cultural Homophily in Grand Slam Doubles

Part 1: ATP Men's Grand Slam Doubles only, 2018–2025 (Olympics excluded throughout — reported separately in Part 2 once this analysis is finalized) · Regression sample: 3,752 team-obs (complete ranking, nationality/language for all 4 players; years since turning pro imputed to 1 for rookies)

1. Data and Measures

Three culture measures capture teammate similarity. **Same nationality** and **same official language** are binary (0/1). **Language proximity** is a continuous 0–1 index of ethnolinguistic closeness (higher = more similar language family).

Regressions use a team-level panel with tournament-by-year fixed effects, round fixed effects, and standard errors clustered by match. Controls in all main specifications: team average doubles ranking, opponent average ranking, top-100 singles indicator (either player), team average years since turning pro, demeaned (*exp_mean_dm*), and its square (*exp_mean_dm_sq*). Culture measures enter one at a time. Rank gap between teammates was tested as an additional control and is insignificant throughout; it is excluded from all main specifications. Nationality and language data are fully resolved in the pipeline (birthplace fallback, surname lookup, Monaco language correction, and `manual_nationality.csv`); zero matches are dropped for nationality.

2. Observation Breakdown

Step	Description	Matches	Change	Team-obs
1	Raw scraped Grand Slam matches (2018–2025, incl. AO/RG/USO 2020; Wimbledon 2020 cancelled)	1,933	—	3,866
2	Drop retirements / walkovers (19 walkovers · 10 retired in set 1 · 18 retired in set 2+)	1,886	−47	3,772
3	Drop: doubles ranking incomplete for ≥1 player (4 of the original 14 were retrieved and restored — see note below)	1,876	−10	3,752
4	Drop: nationality / language missing for ≥1 player (pipeline fully resolves: birthplace, surname lookup, Monaco fix, manual_nationality.csv)	1,876	0	3,752
5	Years since turning pro: 213 obs are rookies (turned pro in/after the tournament year); *exp_mean* imputed to 1 — no obs dropped (these teams have mean rank ≈194 vs ≈122 for the rest; dropping would create selection bias)	—	0	3,752
→ Grand Slams regression sample (Tables 3–4)		**1,876**		**3,752**

Of the 14 matches originally dropped for incomplete doubles ranking, 4 were retrieved: 3 (match_id 6, 35, 49) failed only because Guillermo García-López's accented surname didn't match the unaccented scrape during the ranking merge — his doubles rank was matched in manually; 1 (match_id 913, 2021 Wimbledon) had a genuinely blank partner slot in the raw scrape (Alejandro Davidovich Fokina) — his full profile (rank, nationality, tenure, top-100 history) was reconstructed from public sources and the project's own gravity/ language lookup table. The remaining 10 could not be reliably retrieved and stay dropped.

Table 4 (tiebreak win): 1,182 standard 7-pt tiebreaks × 2 = 2,364 team-obs (928 unique matches), covering sets 1–5. Advantage-set deciders with no breaker played (e.g. 8–6) are not tiebreaks and are excluded entirely — confirmed by the raw tiebreak-score columns being null for all 15 such cases. No genuine 10-point super-tiebreak exists anywhere in this dataset, so there is currently no valid robustness spec to add.

3. Pressure Outcome Counts

Computed on the **same 1,876-match regression sample** used in Tables 3–6 (not a broader pre-filter set), so every number below reconciles exactly with the regression tables.

Outcome	N	Share of matches
Set-1 tiebreak (7-pt)	462	24.6%
Set-2 tiebreak (7-pt)	473	25.2%
Set-3 tiebreak (7-pt)	210	11.2%
Set-4 tiebreak (7-pt)	31	1.7%
Set-5 tiebreak (7-pt)	6	0.3%

Grand Slams regression sample (N=1,876). Sum of the five set-by-set rows: 462 + 473 + 210 + 31 + 6 = **1,182** — this matches Table 4's tiebreak count exactly, since both are computed on the identical sample. Advantage-set deciders with no breaker played (e.g. 8–6, 16–14) are excluded entirely from every row above — a real breaker was never played in those cases, confirmed by the raw tiebreak-score columns being null.

3.1 Match Format: Best-of-3 vs. Best-of-5

Wimbledon played men's doubles as **best-of-5 sets in 2018, 2019, 2021, and 2022** (253 matches played under that rule within the regression sample, of which 101 actually required a 4th or 5th set — the rest finished in straight sets). Every other tournament-year in the Grand Slams — Australian Open, Roland Garros, and US Open in every year, and Wimbledon from 2023 onward — is best-of-3, with no match ever recording a 4th set.

Tournament	Years (best-of-5)	Matches (best-of-5)	Reached 4 sets	Reached 5 sets
Wimbledon	2018, 2019, 2021, 2022	253	101	50
All other tournament-years	—	Best-of-3 throughout	0	0

4. Regression Results

Average marginal effects (AME, dP/dx) from logit models. Each column is a separate model. Standard errors in parentheses, clustered by match. FE: tournament × year, round. All specifications include years-since-turning-pro controls (*exp_mean_dm*, demeaned, and its square). Culture measures enter one at a time.

Table 3 — Outcome: Match Win | N = 3,752 team-obs | AME (d*P*/d*x*)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0377**	+0.0474***	+0.0434***
	(0.0163) p=0.021	(0.0156) p=0.002	(0.0156) p=0.005
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
	(0.0001)	(0.0001)	(0.0001)
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***
	(0.0001)	(0.0001)	(0.0001)
Top-100 singles player	−0.0049	−0.0036	−0.0030
	(0.0315) p=0.876	(0.0315) p=0.909	(0.0315) p=0.924
Team avg. yrs since turning pro	+0.0045***	+0.0043***	+0.0043**
	(0.0017) p=0.008	(0.0017) p=0.010	(0.0017) p=0.011
Yrs since turning pro, squared	−0.0008***	−0.0007***	−0.0007***
	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001
Pseudo-R^2	0.095	0.096	0.096

*** $p<0.01$ ** $p<0.05$ * $p<0.10$. Clustered SE. FE: tournament×year, round. All three culture measures statistically significant at the 5% level (nationality p=0.021, language p=0.002, proximity p=0.005). exp_mean is demeaned (exp_mean_dm = exp_mean − sample mean of 11.90 years since turning pro) and its square is built from the demeaned variable (exp_mean_dm_sq = exp_mean_dm²), applied consistently across every table in this report. This is a pure reparameterization — it leaves the fitted model, the culture AMEs, and every other control's AME unchanged; it only changes what "Team avg. yrs since turning pro" reads as (the marginal effect of tenure evaluated at the sample-mean tenure level rather than at zero tenure). Tenure shows a concave effect: positive linear ($p<0.05$) at the mean, negative quadratic ($p<0.01$). Top-100 singles indicator is not significant: within-GS doubles, having a top singles player does not predict match win once ranking and tenure are controlled.

Table 4 — Outcome: Tiebreak Win | Unit: team × tiebreak | AME (d*P*/d*x*)

Standard 7pt tiebreaks only, any set (1–5) — N = 2,364 team-obs (1,182 tiebreaks, 928 unique matches). Advantage-set deciders with no breaker played are not tiebreaks and are excluded; no genuine 10-point super-tiebreak exists anywhere in this GS dataset, so there is currently nothing valid to add as a robustness spec. Controls: rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq (same as Table 3; exp_mean demeaned against this panel's own estimation-sample mean). FE: tournament×year + round. SE clustered by match.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0298	+0.0075	+0.0019
	(0.0223) p=0.182	(0.0218) p=0.730	(0.0217) p=0.931
Team avg. doubles rank	−0.0003***	−0.0003***	−0.0003***
	(0.0001)	(0.0001)	(0.0001)
Opponent avg. doubles rank	+0.0003***	+0.0003***	+0.0003***
	(0.0001)	(0.0001)	(0.0001)
Top-100 singles player	−0.0663	−0.0634	−0.0632
	(0.0428) p=0.121	(0.0427) p=0.137	(0.0427) p=0.138
Team avg. yrs since turning pro	+0.0026	+0.0022	+0.0022
	(0.0022) p=0.232	(0.0021) p=0.301	(0.0021) p=0.314
Yrs since turning pro, squared	+0.0000	+0.0001	+0.0001
	(0.0003) p=0.929	(0.0003) p=0.866	(0.0003) p=0.854
Pseudo-R^2	0.014	0.013	0.013
N	2,364	2,364	2,364

Unit = one team per tiebreak (2 obs per tiebreak), covering sets 1–5. Culture AMEs are small and insignificant across all three measures — homophily advantage does not extend to within-match tiebreak outcomes. Ranking controls remain strongly significant ($p<0.001$), confirming the model captures real quality differences; tenure controls are not significant in the tiebreak sample.

5. Heterogeneity Analysis

Three sets of interactions are estimated on the main match-win sample (N = 3,752 team-obs), all with the full Table 3 control set (rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq): **Table 5**, culture × surface, testing surface-specific coordination demands; **Table 6**, culture × Hofstede individualism score (team-level, demeaned), testing whether collectivist teams gain more from cultural homophily; and **Table 6a**, culture × years since turning pro, testing whether experienced teams benefit more from cultural alignment.

Hofstede individualism scores (IDV) are taken from Geert Hofstede's official 6-dimension dataset (2015 release, geerthofstede.com). Countries not covered by Hofstede are assigned proxy scores following two tiers: *Tier 1* (direct cultural/linguistic kin, high confidence): BIH→SRB, CYP→GRC, MCO→FRA, MDA→ROU; *Tier 2* (same-region / closest available, moderate confidence): BLR→RUS, BOL→PER, DOM→COL, LBN→Arab composite (IDV=38), TUN→MAR, GEO→TUR, KAZ→RUS, UKR→RUS, UZB→RUS. The team-level score is IC_team = (IDV_p1 + IDV_p2) / 2, demeaned across the GS regression sample (mean = 62.98, SD = 20.85).

Table 5 — Heterogeneity: Culture × Surface | N = 3,752

Two specs, each pooling two surfaces into a single baseline rather than using all three at once. **Spec (a)**: grass vs. rest (hardcourt + clay pooled as baseline). **Spec (b)**: clay vs. rest (hardcourt + grass pooled as baseline). **No**

tournament×year FE (round FE only) — with only two surface groups and no tournament×year FE, the surface main effect is directly estimable rather than absorbed, unlike the collinearity issue that arose with all three surfaces plus tournament×year FE together.

Variable	(1) Same Nationality		(2) Same Language		(3) Lang. Proximity	
	(a) grass	(b) clay	(a) grass	(b) clay	(a) grass	(b) clay
Culture (C) — pooled baseline	+0.0392**	+0.0374**	+0.0438**	+0.0498***	+0.0415**	+0.0422**
	(0.0183) p=0.032	(0.0188) p=0.047	(0.0176) p=0.013	(0.0181) p=0.006	(0.0176) p=0.018	(0.0180) p=0.019
Surface (grass in (a) / clay in (b))	+0.0063	−0.0007	−0.0062	+0.0059	−0.0022	−0.0024
	(0.0148) p=0.671	(0.0142) p=0.962	(0.0206) p=0.765	(0.0188) p=0.753	(0.0207) p=0.914	(0.0194) p=0.903
C × Surface	−0.0082	−0.0003	+0.0140	−0.0106	+0.0075	+0.0038
	(0.0373) p=0.827	(0.0346) p=0.992	(0.0361) p=0.699	(0.0340) p=0.755	(0.0362) p=0.835	(0.0344) p=0.913
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***	−0.0007***	−0.0007***	−0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***	+0.0007***	+0.0007***	+0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0022	−0.0021	−0.0010	−0.0010	−0.0004	−0.0004
	(0.0304) p=0.942	(0.0304) p=0.944	(0.0305) p=0.973	(0.0305) p=0.974	(0.0304) p=0.988	(0.0305) p=0.991
Team avg. yrs since turning pro	+0.0044***	+0.0044***	+0.0043***	+0.0043***	+0.0042**	+0.0042**
	(0.0017) p=0.007	(0.0017) p=0.008	(0.0016) p=0.010	(0.0016) p=0.010	(0.0016) p=0.011	(0.0016) p=0.010
Yrs since turning pro, squared	−0.0007***	−0.0007***	−0.0007***	−0.0007***	−0.0007***	−0.0007***
	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001
Pseudo-R^2	0.095	0.095	0.096	0.096	0.096	0.096
N		3,752		3,752		3,752

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq. FE: round only (no tournament×year FE). SE clustered by match. All surface main effects and interactions are insignificant (p > 0.67 throughout) in both specs. The culture advantage documented in Table 3 does not vary significantly by surface.

Table 6 — Heterogeneity: Culture × Hofstede Individualism (demeaned) | N = 3,752

IC_team_dm = team-average Hofstede IDV score minus the sample mean (62.98). Positive values = more individualist than average. Hypothesis: more individualist teams rely less on in-group bonding → negative θ. Spec 2 only (includes the main IC effect) — the reported specification.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C)	+0.0306*	+0.0401**	+0.0360**
	(0.0166) p=0.065	(0.0162) p=0.014	(0.0162) p=0.026
IC_team_dm	+0.0011*	+0.0008	+0.0008
	(0.0006) p=0.052	(0.0008) p=0.284	(0.0008) p=0.277
C × IC_team_dm (θ)	−0.00049	−0.00003	−0.00001
	(0.00079) p=0.532	(0.00089) p=0.974	(0.00089) p=0.988
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0034	−0.0022	−0.0017
	(0.0316) p=0.914	(0.0317) p=0.943	(0.0316) p=0.958
Team avg. yrs since turning pro	+0.0052***	+0.0050***	+0.0050***
	(0.0017) p=0.002	(0.0017) p=0.003	(0.0017) p=0.004
Yrs since turning pro, squared	−0.0007***	−0.0007***	−0.0007***
	(0.0002) p=0.002	(0.0002) p=0.002	(0.0002) p=0.002
Pseudo-R^2	0.096	0.097	0.097
N	3,752	3,752	3,752

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq. FE: tournament×year + round. SE clustered by match. The culture main effects are consistent with Table 3. The interaction θ is small and insignificant across all three measures ($p > 0.53$), providing no evidence that individualism moderates the cultural homophily advantage. The main IC_team_dm coefficient is positive and marginally significant for same nationality (p=0.052) — teams with higher individualism scores perform somewhat better overall — but is insignificant for the language measures (p>0.27).

Table 6 robustness — dropping Tier-2 Hofstede-proxy countries | N = 3,597

Team-obs involving a player whose IDV score relies on a Tier-2 (moderate-confidence) proxy — BLR, BOL, DOM, GEO, KAZ, LBN, TUN, UKR, UZB — are dropped (155 of 3,752 team-obs), leaving only direct Hofstede matches and Tier-1 (high-confidence) proxies.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C)	+0.0216	+0.0315*	+0.0274*
	(0.0170) p=0.204	(0.0166) p=0.058	(0.0166) p=0.099
IC_team_dm	+0.0007	+0.0001	+0.0001
	(0.0006) p=0.244	(0.0008) p=0.930	(0.0008) p=0.926
C × IC_team_dm (θ)	+0.00008	+0.00083	+0.00085
	(0.00081) p=0.922	(0.00093) p=0.369	(0.00093) p=0.357
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0124	−0.0107	−0.0102
	(0.0329) p=0.707	(0.0330) p=0.745	(0.0330) p=0.756
Team avg. yrs since turning pro	+0.0053***	+0.0051***	+0.0051***
	(0.0017) p=0.002	(0.0017) p=0.003	(0.0017) p=0.003
Yrs since turning pro, squared	−0.0007***	−0.0007***	−0.0007***
	(0.0002) p=0.003	(0.0002) p=0.003	(0.0002) p=0.003
Pseudo-R^2	0.096	0.097	0.097
N	3,597	3,597	3,597

Same controls and FE as Table 6 above (exp_mean_dm carries the full-sample demeaning constant into this subsample rather than being re-centered), estimated on the Tier-2-excluded subsample. Point estimates on C shrink somewhat relative to the full-sample spec (e.g. same nationality: +0.0306 → +0.0216, no longer significant), consistent with the smaller, more homogeneous sample rather than a change in sign or a qualitatively different story. The IC_team_dm main effect for same nationality also loses significance (p=0.052 → p=0.244) once the weaker-confidence proxy countries are dropped, so that result should be treated as sensitive to the Tier-2 approximations. The θ interaction remains small and insignificant throughout, matching the full-sample conclusion that individualism does not moderate the cultural homophily advantage.

Table 6a — Heterogeneity: Culture × Years Since Turning Pro | N = 3,752

exp_mean is demeaned (exp_mean_dm = exp_mean − sample mean of 11.90 years since turning pro, SD = 5.17), consistent with how ic_team_dm is demeaned for Table 6. The main effect of culture (C) is therefore evaluated at the sample-average tenure level, not at zero; the interaction captures how the marginal effect changes per additional year since turning pro relative to that average. exp_mean_dm_sq = exp_mean_dm² (the demeaned quadratic control, used consistently across every table in this report) is included alongside exp_mean_dm, per the blanket rule that all heterogeneity tables carry the full Table 3/4 control set.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C) — mean experience	+0.0377**	+0.0450***	+0.0410***
	(0.0163) p=0.021	(0.0156) p=0.004	(0.0156) p=0.009
C × exp_mean_dm	+0.0003	+0.0064**	+0.0063**
	(0.0032) p=0.925	(0.0032) p=0.045	(0.0032) p=0.049
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0051	−0.0065	−0.0058
	(0.0315) p=0.873	(0.0313) p=0.835	(0.0313) p=0.852
Team avg. yrs since turning pro	+0.0043*	+0.0005	+0.0005
	(0.0022) p=0.054	(0.0025) p=0.845	(0.0025) p=0.848
Yrs since turning pro, squared	−0.0008***	−0.0008***	−0.0007***
	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001
Pseudo-R^2	0.095	0.097	0.097
N	3,752	3,752	3,752

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq (full Table 3 control set). FE: tournament×year + round. SE clustered by match. With exp_mean_dm_sq back in the model, the culture main effects at mean experience are close to Table 3's own estimates (+0.0377/+0.0474/+0.0434) and all significant. The positive and significant interaction for language/proximity (p≈0.045–0.049) implies that advantage grows further with experience above the mean; nationality homophily shows no significant experience gradient (p=0.925).